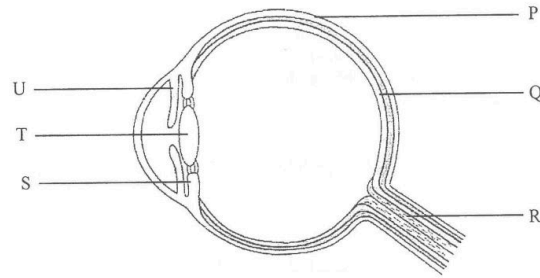


1. 2012Q8

Questions 8 and 9 refer to the diagram below, which shows a section of the human eye:



Which part of the eye is tough and white in colour?

- A. P
- B. Q
- C. R
- D. S

2. 2012Q9

Which of the following parts are made of muscle?

- A. P and S
- B. Q and R
- C. R and T
- D. S and U

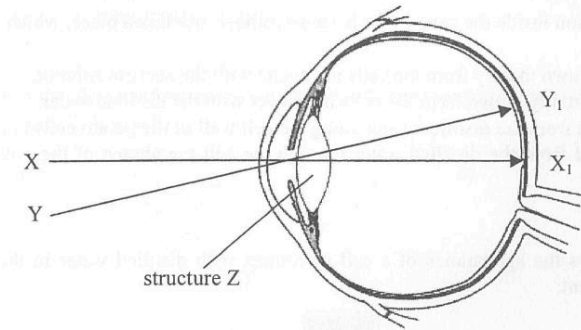
3. 2012Q30

Which of the following correctly describes the importance of phototropism to plants?

- (1) It ensures that the root can get water from the soil.
- (2) It ensures that the root can anchor to the soil for support.
- (3) It allows the shoot to reach a position where there is sunlight.
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

4. 2013Q27

Questions 27 and 28 refer to the diagram below, which shows a section of the human eye and the positions X_1 and Y_1 where the images of two coloured objects X and Y are formed respectively on the retina:



Which of the following correctly describes a person's visual perception in daytime?

- A. X appears brighter because cone cells are found at position X_1 .
- B. X appears sharper because cone cells are found at position X_1 .
- C. Y appears black and white because rod cells are found at position Y_1 .
- D. Y appears blurred because rod cells are found at position Y_1 .

5. 2013Q28

When a person ages, structure Z becomes less elastic. When this happens, a person will probably have difficulty focusing on

- A. a nearby object because structure Z cannot be thickened fully.
- B. a nearby object because structure Z cannot be pulled thin fully.
- C. a distant object because structure Z cannot be thickened fully.
- D. a distant object because structure Z cannot be pulled thin fully.

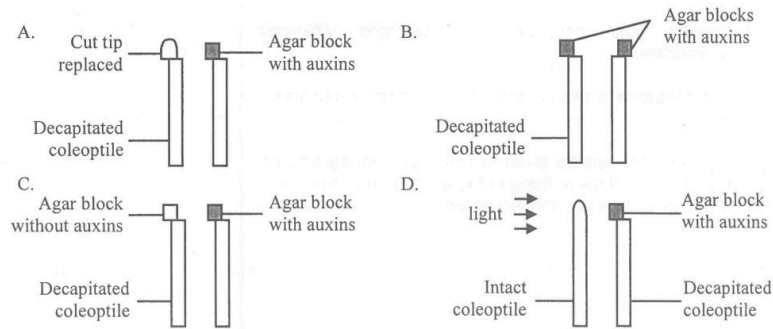
6. 2013Q29

Which of the following combinations correctly matches the structure of the human ear and its function?

- | Structure | Function |
|-----------------|---------------------------------|
| A. ear flap | protecting the ear |
| B. ear drum | amplifying sound waves |
| C. ear bones | transmitting vibrations |
| D. round window | setting the endolymph in motion |

7. 2014Q36

Which of the following pairs of set-ups can be used to test the hypothesis that auxins are growth-promoting substances in oat coleoptiles?



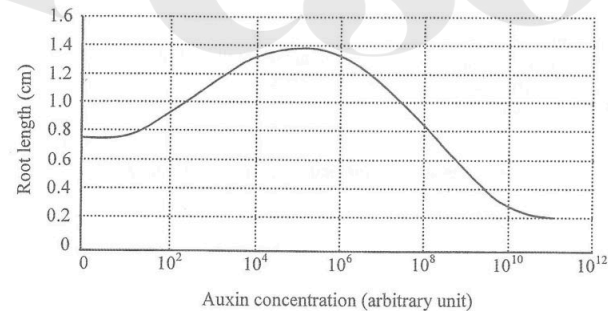
8. 2015Q27

Which of the following combinations correctly shows the conditions of different parts of the eyes when a person is looking at an object moving towards him?

	Lens	Suspensory ligament
A.	becoming thinner	slackening
B.	becoming thinner	tightening
C.	becoming thicker	slackening
D.	becoming thicker	tightening

9. 2015Q29

The graph below shows the average root length of germinating seeds irrigated with different auxin concentrations:

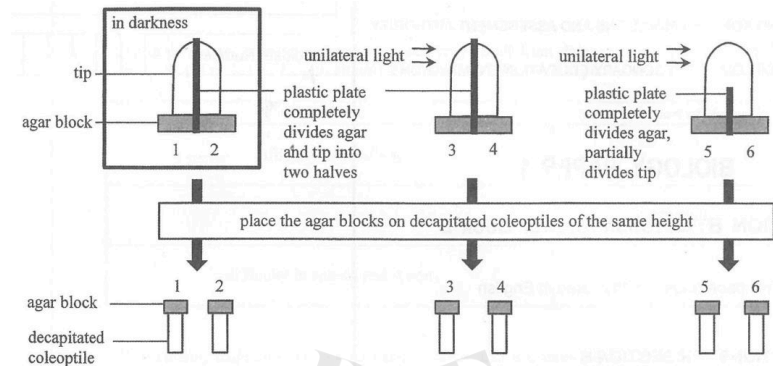


Which of the following can be deduced from the above graph?

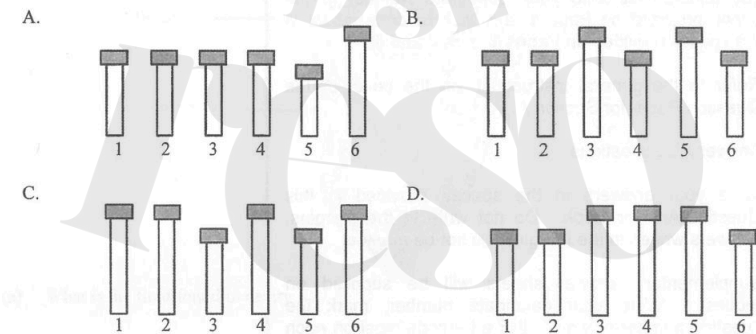
- A. Auxins promote the elongation of the root.
- B. Auxins promote cell division that results in the elongation of the root.
- C. Auxins promote water absorption that results in the elongation of the root.
- D. Different concentrations of auxins result in differences in the extent of root elongation.

10. 2016Q35

Questions 35 and 36 refer to the diagram below, which shows an experiment that collects auxins from the tip of coleoptiles under different conditions. After that, the agar blocks are placed on decapitated coleoptiles of the same height:

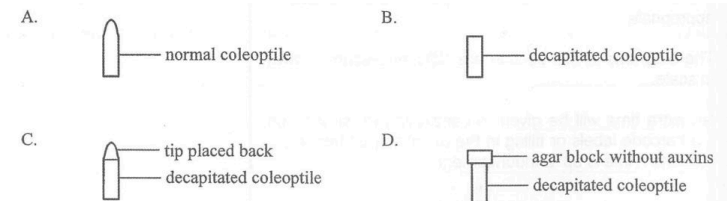


Which of the following correctly shows the growth response of the decapitated coleoptiles?



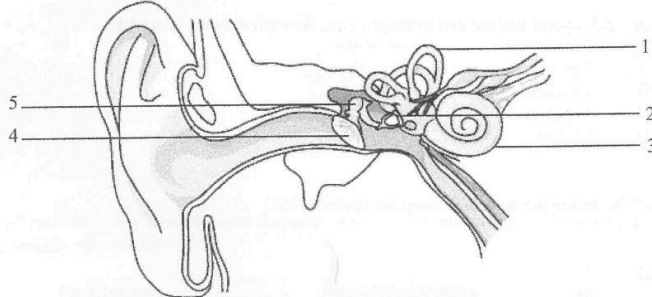
11. 2016Q36

Which of the following can be used as a control set-up for the above experiment?



12. 2017Q25

Questions 25 and 26 refer to the diagram below, which shows the structures of a human ear:

Which of the following structures is *not* involved in hearing?

- A. 1
- B. 2
- C. 3
- D. 4

13. 2017Q26

Which of the following structures will vibrate when there are sound waves?

- A. 1 and 3 only
- B. 2 and 3 only
- C. 2, 3 and 4 only
- D. 2, 4 and 5 only

14. 2017Q27

Which of the following combinations correctly matches the problem of short sightedness and its correction?

- | Problem | Correction |
|---|-------------------|
| A. image focused behind the retina | wear concave lens |
| B. image focused behind the retina | wear convex lens |
| C. image focused in front of the retina | wear concave lens |
| D. image focused in front of the retina | wear convex lens |

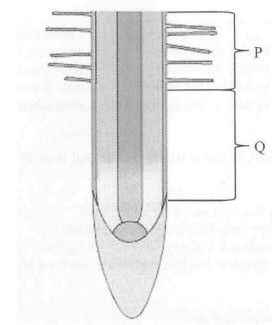
15. 2017Q28

Which of the following correctly explains why our eyes feel tired if we have been reading a book for a long time?

- A. The ciliary body has contracted for a long time.
- B. The photosensitive cells have been over-stimulated.
- C. The lens has maintained a curved state for a long time.
- D. The suspensory ligament has been under tension for a long time.

16. 2017Q31

The diagram below shows a low power drawing of the cut section of a root:

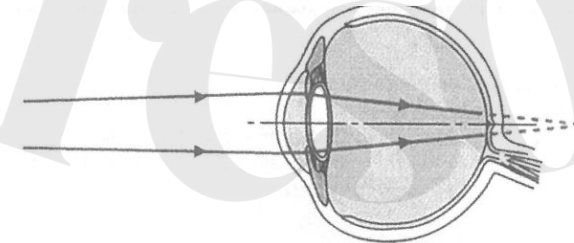


Which of the following correctly describes the relationship of auxin and different regions of the root?

- A. It affects cell division at P.
- B. It affects cell division at Q.
- C. It affects cell elongation at P.
- D. It affects cell elongation at Q.

17. 2018Q7

Below is a ray diagram of a common eye defect:



Which of the following combinations correctly identifies the eye defect and type of lens to be worn to remedy it?

- | | Eye defect | Type of lens |
|----|-------------------|---------------------|
| A. | Long-sightedness | Convex lens |
| B. | Long-sightedness | Concave lens |
| C. | Short-sightedness | Convex lens |
| D. | Short-sightedness | Concave lens |

18. 2019Q16

Peter wanted to throw a crumpled paper into the rubbish bin in a dim room. He found that he could see the bin more clearly if he tried to focus on objects right next to the bin. Which of the following statements about the image formation of the bin is correct?

- A. Image of the bin is formed on the yellow spot where there are cone cells only.
- B. Image of the bin is formed on the yellow spot where there are more cone cells than rod cells.
- C. Image of the bin is formed on the periphery of the retina where there are rod cells only.
- D. Image of the bin is formed on the periphery of the retina where there are more rod cells than cone cells.

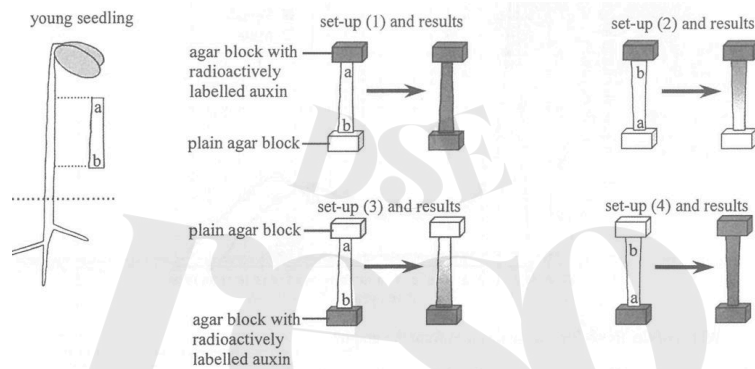
19. 2019Q17

Wearing contact lenses for too long will have an adverse effect on the eyes because this decreases the amount of oxygen reaching the eyes. Which of the following eye structures is most affected in this case?

- A. iris
- B. lens
- C. cornea
- D. sclera

20. 2019Q28

To investigate the transport of auxins in the stem of a young seedling, a plain agar block and an agar block soaked in radioactively labelled auxins were prepared and placed at different ends of a cut stem, as shown in the diagram below. The relative amounts of radioactivity in different parts are shown as different intensities of shaded area.

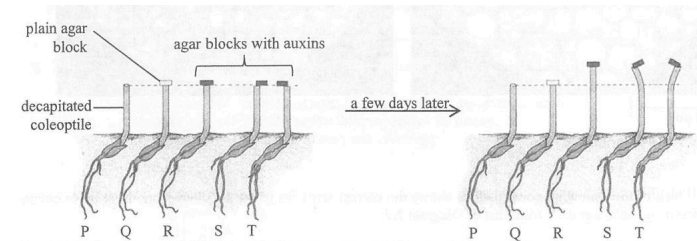


Which of the following conclusions can be drawn from the above results?

- (1) Transport of auxins in the stem is not affected by gravity.
 - (2) Transport of auxins in the stem involves an active process.
 - (3) Transport of auxins in the stem mainly takes place from a to b.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

21. 2020Q27

Questions 27 and 28 refer to the diagram below, which shows an investigation into the effect of auxins on the growth of plant shoots:



Which of the following serve(s) as control set-up(s) in this investigation?

- A. P only
- B. R only
- C. P and Q only
- D. Q and R only

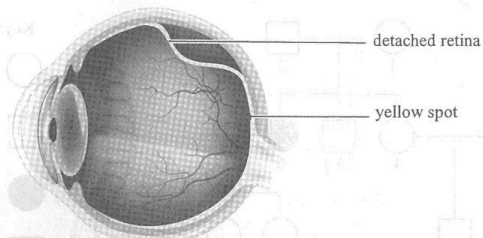
22. 2020Q28

Which of the following conclusions can be drawn based on the results of the investigation?

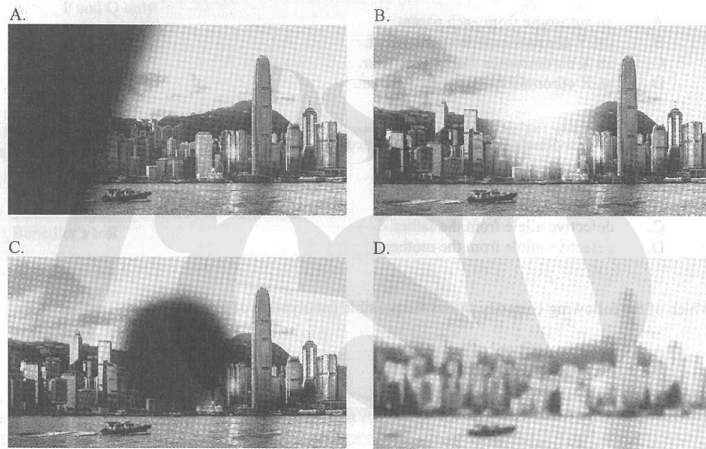
- A. The tip of the coleoptile produces auxins.
- B. Auxins stimulate the growth of coleoptiles.
- C. The coleoptile shows positive phototropism.
- D. Coleoptiles stop growing when the tip is removed.

23. 2021Q18

The diagram below shows an early stage of an eye defect:

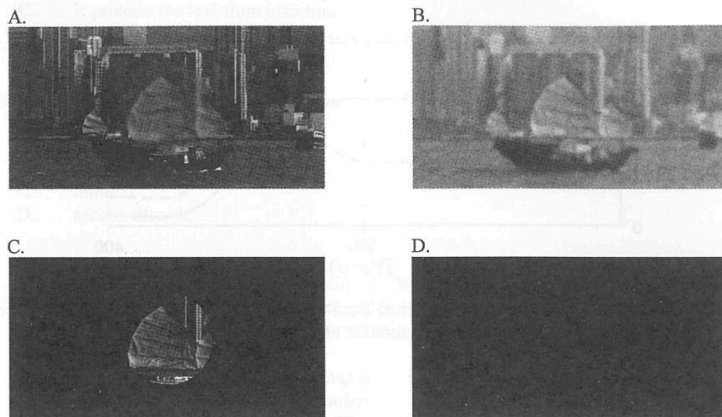


If a person suffers from this eye defect, which of the following diagrams is the most likely vision perceived by this person?



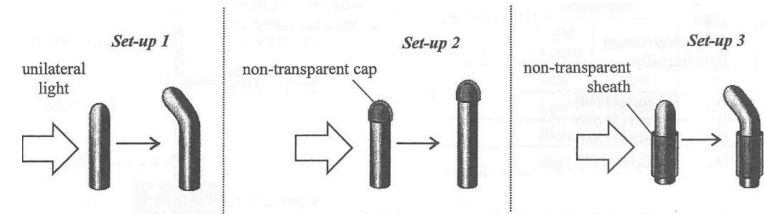
24. 2022Q24

During aging, some people may suffer from an eye defect which is caused by the degeneration of light-sensitive cells in the yellow spot. Which of the following diagrams is the most likely vision perceived by a person suffering from this eye defect?



25. 2022Q25

Darwin performed some experiments on the study of the phototropic response of coleoptiles of young seedlings. The coleoptile in each set-up was exposed to a unilateral light source and the appearance of the coleoptile after three days are shown below:



Which of the following can be deduced from Darwin's experiment?

- (1) Auxins is produced from the tip of the coleoptile.
 - (2) The coleoptiles show positive phototropic response.
 - (3) The tip of the coleoptile detects the direction of light.
- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

26. 2023Q19

Blockage of the Eustachian tube will directly affect the functioning of the

- A. cochlea.
- B. eardrum.
- C. ear bones.
- D. oval window.

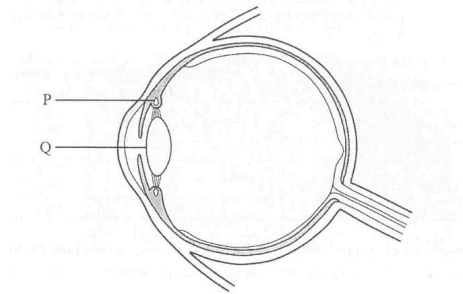
27. 2024Q22

Which of the following combinations correctly compares pupil reflex in humans and phototropism in plants?

- | | pupil reflex | phototropism |
|---------------------------|---------------------|---------------------|
| (1) stimulus | light intensity | direction of light |
| (2) location of receptors | retina | shoot tip |
| (3) effector | pupil | shoot |
- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

28. 2024Q27

Questions 27 and 28 refer to the diagram below, which shows a section of a human eyeball:



Ken and Jane are looking at a bird flying away from them. Which of the following combinations correctly describes the conditions of structures P and Q of their eyeball

- | | <i>Structure P</i> | <i>Structure Q</i> |
|----|--------------------|--------------------|
| A. | relaxing | becoming thinner |
| B. | relaxing | becoming thicker |
| C. | contracting | becoming thinner |
| D. | contracting | becoming thicker |

29. 2024Q28

When the bird flies away to a certain distance, Ken can still see it clearly but Jane cannot. Ken has normal eyesight. Which of the following combinations correctly describes Jane's condition

- | | <i>Eyeball</i> | <i>Position of image</i> |
|----|----------------|--------------------------------|
| A. | too long | focused behind the retina |
| B. | too long | focused in front of the retina |
| C. | too short | focused behind the retina |
| D. | too short | focused in front of the retina |

30. 2024Q30

Questions 30 and 31 refer to an investigation about the effects of auxins on the growth of shoots. 10 mm sections of shoots were obtained from a number of seedlings. Auxin solutions of different concentrations were prepared. Three shoot sections were put into each solution for two days. The results are shown in the table below.

Auxin concentration (ppm)	Length of the shoot section after 2 days (mm)		
	Shoot 1	Shoot 2	Shoot 3
0	15.0	14.5	15.1
0.1	32.5	32.4	32.2
1	37.1	37.2	10.1
10	24.0	23.9	23.8
100	12.5	12.5	13.0
1000	10.0	9.8	10.3

Based on the above results, which of the following is the lowest auxin concentration that inhibits the growth of the shoots?

- A. 0.1 ppm
- B. 10 ppm
- C. 100 ppm
- D. 1000 ppm

31. 2024Q31

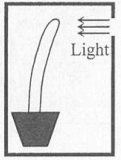
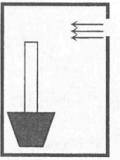
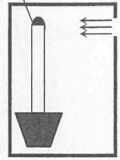
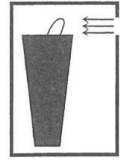
Which of the following best explains the result of Shoot 3 in 1 ppm auxin concentration solution?

- A. This datum is anomalous.
- B. It is an error which should be deleted.
- C. It is due to variations in individual differences.
- D. The shoot section is likely taken from the region of cell differentiation.

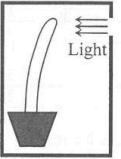
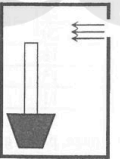
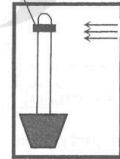
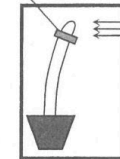
LQ

1. 2013Q7

In 1880, Darwin conducted an experiment to investigate the phototropism of plants. He placed some coleoptiles in dark boxes, each with a hole at one side to allow light to pass through. The results after various treatments of coleoptiles are shown in the diagrams below.

Set-up	I	II	III	IV
Treatment of the coleoptile	Intact coleoptile	Tip removed	Opaque cap placed on the tip	Buried in soil with tip exposed
Result	 Growth with bending	 No growth and no bending	 Growth without bending	 Growth with bending

- a) From the results of the experiment, which part of the coleoptile is responsible for detecting unilateral light? Support your answer with reasons. (3 marks)
- b) Explain why it is necessary to have set-up III in the experiment. (1 mark)
- c) In 1913, Boysen-Jensen performed some other experiments to study the nature of the signal transmission involved in phototropism. The diagram below shows his experimental set-ups:

Set-up	A	B	C	D
Treatment of the coleoptile	Intact coleoptile	Tip removed	Cut tip placed on a mica block	Cut tip placed on an agar block
Result	 Growth with bending	 No growth and no bending	 No growth and no bending	 Growth with bending

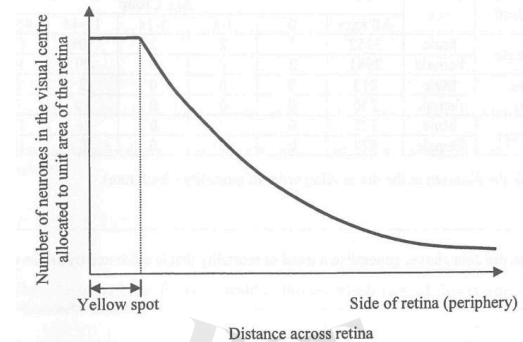
What conclusion can be drawn from Boysen-Jensen's experiment? (3 marks)

- d) Which of the following statements about the nature of science are demonstrated in the above historical events? Put a '✓' in the space next to the statement and provide an explanation. The first one is an example for your reference. (4 marks)

Statement		Evidence from the historical events
Doing science requires creativity and imagination.	✓	Both Darwin and Boysen-Jensen used innovation and imagination to design their experiments.
Science is socially and culturally embedded.		
Science is based on evidence.		
Science knowledge is tentative and dynamic.		

2. 2014Q10a

The visual center of the brain is responsible for processing nerve impulses from the retina. The number of neurones in the visual centre allocated to receiving impulses from a unit area of the retina is shown below (only half of the retina is shown):



- a) With reference to the types of photoreceptor cells and their distribution on the retina, explain why more neurones in the visual centre are allocated to unit area of the yellow spot. (4 marks)

3. 2015Q1

For each of the ear parts listed in column 1, select from column 2 one phrase that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1

Ear bones

Eustachian tube

Cochlea

Column 2

A. Transmitting vibrations

B. Transmitting sound waves

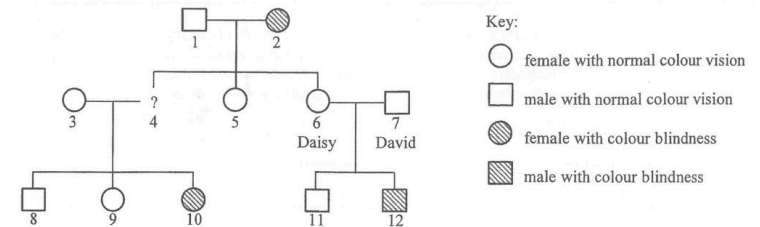
C. Converting sound waves to vibrations

D. Converting vibrations to nerve impulses

E. Equalizing the air pressure on either side of the ear drum

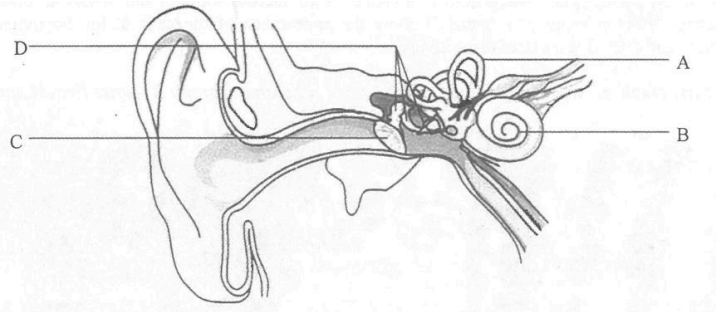
4. 2016Q10a

Color blindness is an X-linked recessive genetic disorder. The pedigree below shows the inheritance of colour blindness in a family:



- a) Colour blindness is due to the abnormal development of photoreceptors. State the relevant type of photoreceptors and the location inside the eyeball where these photoreceptors are most abundant. (2 marks)

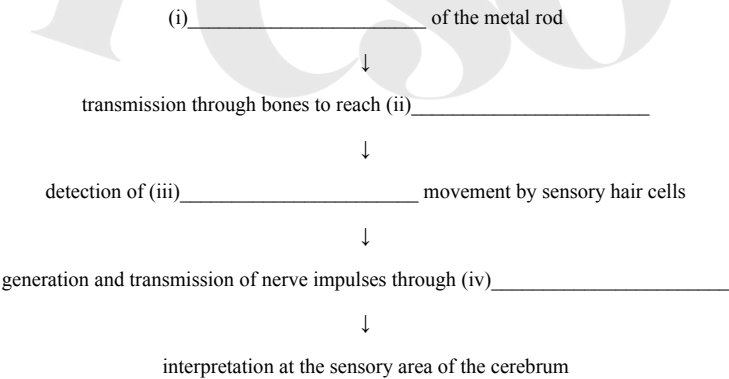
5. 2018Q1
The diagram below shows the human ear and its associated structures:



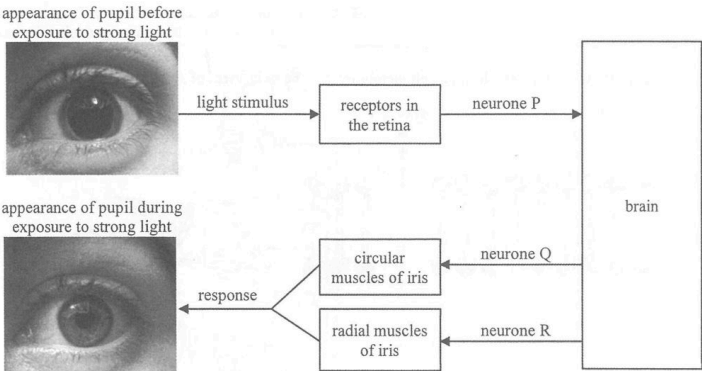
a) The table below lists two types of hearing loss. Using the label(s) in the above diagram, indicate which structure(s) is / are most likely to be defective in each case. (2 marks)

	Type of hearing loss	Structure
X	Damage to sensory hair cells	
Y	Failure of sound conduction	

b) Ludwig van Beethoven, a famous 18th century composer, suffered from type Y hearing loss (4 marks) in his 20s and became deaf in his 40s. Some records say that he could hear music through his jawbone and skull by biting on a metal rod attached to his piano. Based on the structures and functions of human ears, complete the following flow chart to show the major steps involved in his method of hearing music.



6. 2020Q3ai, b
The diagram below shows a neural pathway involved in the coordination of pupil size when a human eye is exposed to strong light:



- a) With reference to the above neural pathway.
- i) State the **two** types of receptors located in the retina. (1 mark)
- b) Describe how the two sets of iris muscles work together to bring about the change in the pupil size shown above. State the significance of this response. (4 marks)

7. 2023Q2a
All cells are derived from stem cells. They undergo differentiation in which the cells change in form and shape which enable them to perform specialised functions.

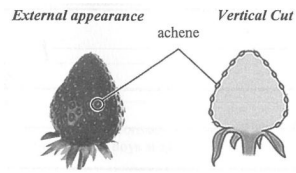
a) It is found that the lens of the eye is composed of cells without organelles. If the organelles of these cells had not been degraded during differentiation, describe how the functioning of the lens would have been affected. (2 marks)

8. 2023Q6a
Color blindness is an inherited disorder due to defective functioning of the cone cells in the retina. There are many types of color blindness. For example, people with red-green color blindness fail to distinguish between red and green colors, while those with total color blindness experience total loss of color vision.

a) Based on the functioning of cone cells, suggest why the condition of red-green color blindness is different from that of total color blindness. (1 mark)

9. 2023Q9

The diagram below shows the external appearance of a strawberry and its vertical cut. The achenes found on the surface of the strawberry are the fruits:



a) An investigation into the role of achenes in the development of a strawberry was carried out as shown below:

Treatment	Relative size and appearance of the strawberry	
	Day 1	Day 20
1. Achenes remained intact		
2. All achenes were removed on Day 1		
3. All achenes were removed on Day 1 and the strawberry was then regularly sprayed with auxins.		

i) Complete the following table to show what deduction can be made by comparing results of the following treatments: (3 marks)

Treatment	Deduction
1 verses 2	
2 verses 3	
1 versus 3	

ii) Based on the results, suggest **one** hypothesis for the enlargement of the strawberry. (1 mark)

iii) Study another treatment as follows:

Treatment	Relative size	
	Day 1	Day 20
4. Achenes were removed from the lower part of the strawberry on Day 1.	Achenes remained on the upper part Achenes removed from the lower part	

In terms of experimental design, what is the advantage of Treatment 4 as compared to Treatments 1 and 2? (1 mark)

b) Give **one** example of a growth response induced by auxins and state its significance to plants. (2 marks)

Ans

2012Q8: A (81%)	2012Q9: D (87%)	2012Q30: C (35%)	2013Q27: B (64%)
2013Q28: A (55%)	2013Q29: C (62%)	2014Q36: C (77%)	2015Q27: C (72%)
2015Q29: D (91%)	2016Q35: A (42%)	2016Q36: D (64%)	2017Q25: A (79%)
2017Q26: D (73%)	2017Q27: C (71%)	2017Q28: A (66%)	2017Q31: D (65%)
2018Q7: A (74%)	2019Q16: D (56%)	2019Q17: C (59%)	2019Q28: B (45%)
2020Q27: C (61%)	2020Q28: B (84%)	2021Q18: A (68%)	2022Q24: A (52%)
2022Q25: C (56%)	2023Q19: B (51%)	2024Q22: A (45%)	2024Q27: A (64%)
2024Q28: B (61%)	2024Q30: C (36%)	2024Q31: D (19%)	

2013Q7

- 7 (a) • the tip is responsible for detecting the unilateral light (1)
 • because if it is removed / blocked from the stimulus, no bending growth movement occurs (1) (3)
 • on the other hand, the same response is produced even if the lower part of the coleoptile is covered in soil (1)
- (b) • to show that the failure to produce a response in set-up II is not due to the effect of injury / damage when the tip is removed (1) (1)
- (c) • some substances are produced from the tip of the coleoptiles (1)
 • which can diffuse through the agar block (1) to reach the lower part of the coleoptile (3)
 • and exert effect / lead to bending growth at the lower part of the coleoptile (1)
- | | | | |
|---|-------|---|-----|
| (d) Science is based on evidence. | ✓ (1) | Both Darwin and Jensen used the results from their experiment to develop their understanding about phototropism in plants (1) | (4) |
| Science knowledge is tentative and dynamic. | ✓ (1) | Darwin's work only provided some understanding about phototropism and Jensen's results helped develop further the science knowledge (1) | |

11 marks

2014Q10a

- 10 (a) • yellow spot contain a high density of cone cells (1)
 • and there are three types of cones for color perception (1),
 • as a result, there are more sensory nerve impulses coming from the yellow spot (1) (4)
 • hence more neurones are allocated to analysis of the nerve impulses from the yellow spot (1)

2015Q1

- 1 A (1)
 E (1)
 D (1)

3 marks

2016Q10a

- 10 (a) • cone cells (1) (2)
 • they are concentrated at the yellow spot (1) of the retina

2018Q1

- 1 (a) • B (1) (2)
 • C and D (1)
- (b) (i) vibrations (1)
- (ii) oval window / cochlea (1) (4)
- (iii) lymph / fluid (1)
- (iv) auditory nerve (1)

6 marks

2020Q3ai, b

- 3 (a) (i) • rod cells and cone cells (1) (1)
- (b) • circular muscles of iris contract while radial muscles of iris relax (1)
 • leading to the constriction of the pupil (1) (4)
 • to reduce the amount of light entering the eyes (1)
 • this prevents overstimulation of rod cells and cone cells (1)

2023Q2a

- 2 (a) Concept for mark award (2)
 • description of how the presence of organelles affects the lens (1)
 • description of how it affects the image formed (1)
- e.g.
 • light is scattered by the presence of organelles / some light is blocked by the presence of organelles / the lens will become unclear (1)
 • Images formed on the retina become unclear / blurred / light cannot focus correctly on the retina (1)

2023Q6a

- 6 (a) • one type of cone cell / the cone cells for detecting gree spectrum do not function in red-green colour blindness while all three types of cone cells do not function in total blindness (1) (1)

2023Q9

- | Treatment | Deduction |
|------------|--|
| 1 versus 2 | • it is shown that achenes are responsible for the enlargement of the strawberry (1) |
| 2 versus 3 | • it is shown that auxin is the plant hormone that causes enlargement of the strawberry (1) (3) |
| 1 versus 3 | • no conclusion / deduction can be reached by comparing the results of set-ups 1 and 3 as there is more than one variable in the two set-ups (1) |
- (ii) • achenes produce auxins, which cause the enlargement of the strawberry during its development (1) (1)
- (iii) • it eliminates the individual difference between the strawberries in set-ups 1 and 2 (1) (1)
- (b) Any **one** set of the following:
 • positive phototropic response in shoot (1) to ensure the shoot receives light for photosynthesis (1)
 • negative phototropic response in root (1) to ensure the root grows into the soil (1) (2)